CRISTIAN ROBLES

US Citizen ⋄ Pasadena ⋄ (626) · 689 · 3842 ⋄ cristian.robles.762@my.csun.edu ⋄ https://robles-c.github.io/

EDUCATION

California State University, Northridge May 2026
Master of Science in Computer Engineering *Northridge, CA*
Overall GPA: 3.31

SKILLS

Embedded: C/C++, Python, SPI/I2C/UART/CAN, RTOS, Yocto Embedded Linux, Device Drivers, NASA F´ Flight Software, Microcontroller Bring-Up, SWD/JTAG

FPGA Design: Verilog, PolarFire SoC, Xilinx FPGA, HDL Sensor Interfaces, Bus Protocol Design, Vivado, Libero SoC

Hardware Testing & Electronics: PCB Design, SMD/TH Soldering, Power Electronics, Oscilloscope, Logic Analyzer, DMM, Signal Integrity Debugging, Hardware Validation

Robotics: PID, Kalman Filters, Motor Control, Odometry, ROS, SolidWorks, 3D Printing, Rapid Prototyping

WORK EXPERIENCE

Hidonix Inc. August 2025 – Present
Firmware Engineer *Santa Monica, CA*

- Contribute across the full robotics stack, including embedded firmware, embedded Linux development, and mechanical design.
- Develop low-level firmware for motors, sensors, CAN devices, and custom electronics; perform hardware bring-up, debugging, and system validation.
- Design and integrate mechanical components and assemblies supporting actuators, linkages, and robotic mechanisms.
- Build and maintain embedded Linux environments for robot compute modules, including hardware interfacing and driver-level integration.
- Lead rapid prototyping efforts and collaborate with cross-disciplinary teams to deliver reliable robotic systems from concept to deployment.

PROJECTS

Graduate Project: F´-Based Tank Robot With Deployable Drone March 2025 – Present

- Designing a full robotics system as a graduate project, centered on a rover-style tank platform controlled using NASA's F Prime flight software framework.
- Developing a deployable ESP32-based drone intended to communicate with the rover over UDP for real-time telemetry.
- Migrating the rover's main compute module from a Raspberry Pi to the Microchip PolarFire SoC to achieve deterministic real-time behavior and support future expansion.
- Implementing FPGA-based hardware acceleration by creating custom logic to decode encoders, IMUs, ToF sensors, and other high-rate peripherals directly in hardware.
- Offloading repetitive sensor polling and control-loop tasks from the CPU into memory-mapped FPGA registers, improving timing precision, reducing CPU load, and enabling higher-bandwidth control.
- Long-term objective: a flight-software-inspired rover–drone system with robust communication, FPGA-accelerated sensing/actuation, and reliable real-time robotics performance.